Project Design Phase-II
Data Flow Diagram & User Stories

Date	4 March 2026
Team ID	LTVIP2026TMID_HEART
Project Name	Heart Disease Analysis Using Tableau Visualization
Maximum Marks	4 Marks

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of system requirements graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 — Context Diagram

DFD LEVEL 0 — Heart Disease Analysis System

[User] ——————▶ [Heart Disease Analysis System] ——————▶ [Stakeholders]

Inputs: Heart_new2.csv (4,500 records, 18 columns) | Outputs: Interactive Tableau Dashboards & Story

DFD Level 1 — System Flow

STEP 1: User Receives Dataset (Heart_new2.csv)

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STEP 2: Import Dataset into MySQL Database

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STEP 3: Is Data Cleaned? → No → Clean via Python/Excel → Return to Step 2

▼ **Yes**

STEP 4: Create Visualizations in Tableau Sheets (10 Charts)

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STEP 5: Heart Disease Risk Analysis Output (Dashboard + Story)

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STEP 6: Publish Dashboard to Tableau Public

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STEP 7: Stakeholders View Published Dashboard [Dr. Sharma | Ramesh | Anita]

User Stories

User stories define the functional expectations of the Heart Disease Analysis Tableau dashboard from the perspective of each key stakeholder. Each story follows the format: As a [user type], I can [action], so that [benefit].

User Type	Functional Requirement	Story No.	User Story	Acceptance Criteria	Priority	Release
Cardiologist (Dr. Sharma)	Explore patient risk profiles	USN-1	As a cardiologist, I can view heart disease cases filtered by age, gender, and BMI	I can filter patients and view risk distribution clearly across all demographics	High	Sprint-1
Cardiologist (Dr. Sharma)	Compare lifestyle factors	USN-2	As a cardiologist, I can compare smoking and alcohol impact on heart disease cases	I can identify the highest risk lifestyle combinations using stacked bar charts	High	Sprint-1
Health Analyst (Ramesh)	View disease co-occurrence	USN-3	As a health analyst, I can see stroke, diabetes, and kidney disease alongside heart disease	I can view co-occurrence patterns clearly segmented by age group	High	Sprint-1
Individual (Anita)	Filter by personal health factors	USN-4	As an individual, I can filter by BMI range and age category to assess my personal risk	I can view only my relevant age and health group in the dashboard	Medium	Sprint-2
Individual (Anita)	View general health vs heart disease	USN-5	As an individual, I can view how general health ratings correlate with heart disease	I can compare Excellent, Good, Fair, and Poor health outcomes on bubble chart	Medium	Sprint-2
Admin / Analyst	Upload or update dataset	USN-6	As an analyst, I can update patient data into MySQL and refresh Tableau connection	Dashboard reflects new data without requiring complete redesign or rebuild	Low	Sprint-2

Summary

The Data Flow Diagram illustrates how raw heart disease data from Heart_new2.csv is ingested into MySQL, cleaned, transformed into 10 Tableau visualizations, and published to Tableau Public. The six user stories capture the complete feature set required by all three stakeholders — ensuring the dashboard delivers on clinical, policy, and personal health monitoring needs.